Freight Rate Prediction

Machine Learning Engineer Assessment

48,000	136.94	5.72%
labeled loads	holdout MAE ($)	holdout WAPE

1. Objective and data

The objective is to predict **posted_rate** for each load. The supplied development data contains 48,000 labeled rows dated January through October 2025. The separate November-December file contains 12,000 unlabeled loads. The identifier **load_id** is retained only for output matching and excluded from model features.

A quick audit found 300 missing weight values and 374 missing market_index values in the labeled data. The main validation file has 165 missing weights and 249 missing market_index values. Rates range from $57.22 to $25,533.00, with a median of $2,030.76. Distance has a strong raw association with posted rate (Pearson correlation about 0.91); this is a useful screening observation, not a complete explanation of rate behavior.

2. Split and model selection

I used a forward-in-time holdout: train on January 1-August 31 (38,477 rows) and validate on September 1-October 31 (9,523 rows). This better reflects forecasting later loads than a random split. I compared an equipment-specific median rate-per-mile baseline, a Ridge regression, and a HistGradientBoostingRegressor. Model selection used holdout MAE; RMSE and WAPE are also reported.

Model	MAE ($)	RMSE ($)	WAPE	Mean error ($)
Median rate per mile by equipment	229.10	670.66	9.58%	40.93
Ridge regression	171.98	647.84	7.19%	-111.56
HistGradientBoosting (selected)	136.94	636.67	5.72%	-49.31

The selected model achieved a holdout MAE of $136.94, RMSE of $636.67, and WAPE of 5.72%. RMSE is much larger than MAE, showing that a smaller number of large misses remain; MAE is therefore the primary selection metric. The mean prediction error was -$49.31, indicating modest underprediction on this holdout.

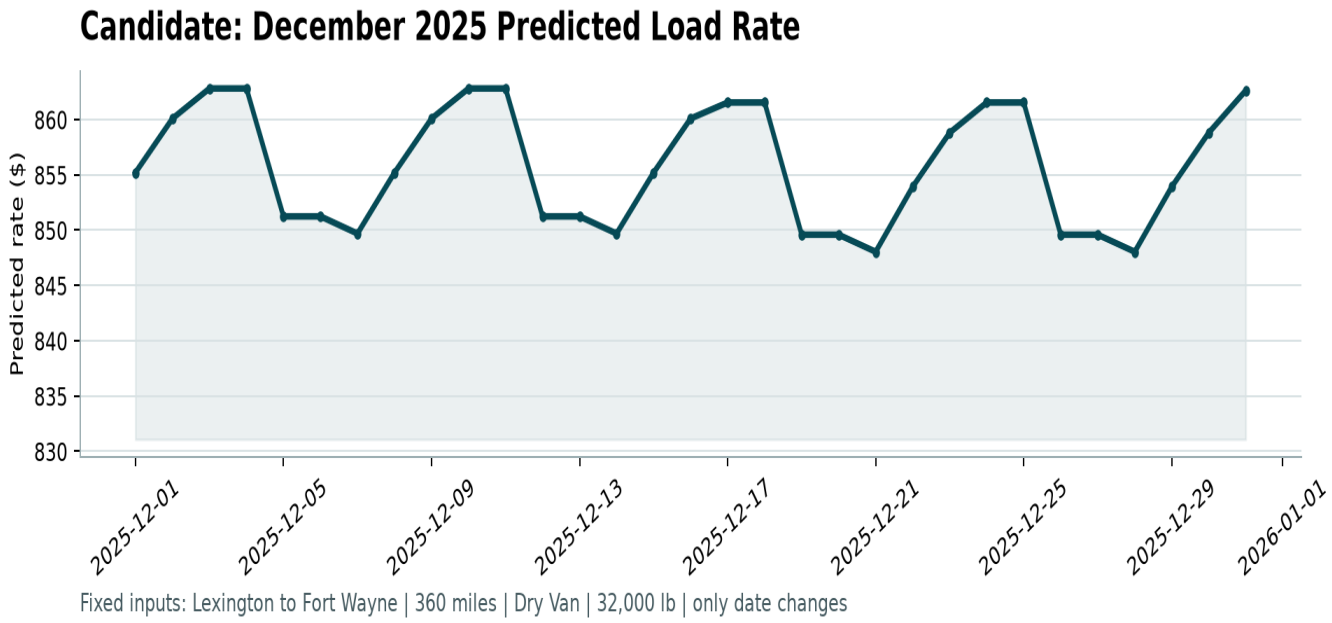
Implementation and December Scenario

3. Feature and missing-data handling

Pickup, delivery, and equipment are one-hot encoded. Numeric inputs include coordinates, distance, weight, market index, quote signal, and calendar features derived from date (month, day of year, day of week, weekend flag, seasonal sine/cosine, and ordinal date). Numeric medians are learned only within each fitted training pipeline, and missingness indicators are included. Unseen categories are ignored safely. The model is refit on all 48,000 labeled rows after validation and then used for final predictions.

The December chart input intentionally provides only pickup, delivery, distance, equipment, weight, and date. Coordinates, market index, and quote signal are unavailable in that scenario; the fitted numeric imputer supplies training medians for those absent columns. Therefore, the chart is a date-varying prediction for the supplied fixed load, not a forecast conditioned on daily market signals.

4. December 2025 predicted-rate chart



5. Reproduction and limitations

From the project folder, install requirements, run **python run_assessment.py**, then run the scorer command shown in README.md. The script writes validation_predictions.csv, december_chart_inputs_completed.csv, and validation_metrics.json. This example uses one two-month forward holdout and limited tuning; a production-quality study would add multiple expanding-window checks and residual analysis. The measured holdout is evidence for model choice, not a guarantee of November-December accuracy.